# GRATICULE INSTRUMENTS LTD

Applied AI research and production

# Development Cost Statement (Founder Basis)

Applied AI platform development: model engineering, evaluation, interface and deployment

Statement of the effort, infrastructure and third-party consumption applied to the design, build, verification and evaluation of the platform, covering the period to 15 October 2026.

*Basis: Founder-rate basis, at cost*

| £8,620 | £289 | 23 | £8,909 |
|---|---|---|---|
| Engineering effort | Infrastructure, data and API consumption | Engineer days | Total cost of development |

## Basis of preparation

Figures shown as MEASURED are verified against the delivered repository and are reproducible: code volume, artefact sizes, evaluation corpus size and verification results.

This statement is prepared on a founder basis. The principal engineer is charged at cost rather than at commercial market rate, and the effort recorded is direct build time rather than a fully loaded commercial engagement.

The scope delivered is identical to the commercial statement. The difference in total reflects the rate and effort basis, not a reduced deliverable.

This is the appropriate basis where the work was carried out by the founding team and the purpose is to evidence investment made rather than to raise an invoice at market rates.

*The entity is in the process of incorporation. No VAT is chargeable at this time and this statement does not constitute a demand for payment.*

# 1. Scope delivered

*All figures in this section are MEASURED from the repository.*

## 1.1 Production code

| Component | Modules | Lines |
|---|---|---|
| Model, inference and API layer (Python) | 25 | 2,388 |
| Interface layer (JavaScript, JSX, CSS) | 16 | 1,438 |
| **Total** | **41** | **3,826** |

## 1.2 Trained artefacts and deployment assets

| Artefact | Size | Status |
|---|---|---|
| model.pth (ResNet50 backbone, 3-class head) | 94.36 MB | Delivered |
| LungDisease_Classifier.pth (LC25000CNN, 4,750,885 parameters) | 19.01 MB | In service |
| lung_classifier.onnx (browser inference graph) | 19.01 MB | In service |
| WebAssembly inference runtime, vendored and pinned | 14.24 MB | In service |
| Technical and funding documentation | 0.03 MB | Delivered |

## 1.3 Verification and evaluation

| Activity | Result | Basis |
|---|---|---|
| Numerical parity, PyTorch against ONNX Runtime | 3.8e-06 maximum absolute difference | Measured |
| Labelled evaluation corpus | 120 images, 40 per class | Measured |
| Label mapping determination | All 6 permutations evaluated, best identified | Measured |
| End-to-end smoke test | 6 of 6 checks passed | Measured |
| Three-class accuracy on held-out data | Not yet established | Outstanding |

*The evaluation corpus is drawn from the same public dataset used for training and is therefore not a held-out result. Establishing a held-out split is the first milestone of the funded programme.*

# 2. Engineering effort applied

*MODELLED. Effort is derived from the delivered scope in section 1 and should be reconciled to timesheets.*

| Role | Day rate | Days | Amount |
|---|---|---|---|
| Principal Machine Learning Engineer (founder, at cost) | £420 | 12 | £5,040 |
| Software Engineer, front end | £350 | 6 | £2,100 |
| Research Scientist, evaluation and validation | £320 | 3 | £960 |
| Technical Author, documentation | £260 | 2 | £520 |
| **Total engineering effort** | | **23** | **£8,620** |

### 2.1 Effort distribution across the delivered scope

| Workstream | Description | Days (full basis) |
|---|---|---|
| Discovery and architecture | Problem definition, data strategy, architecture and deployment model selection. | 6 |
| Model integration | Checkpoint reconstruction, architecture matching, inference layer, ONNX export. | 10 |
| Front end | Clinical dashboard, upload pipeline, results panel and explainability viewer. | 12 |
| Explainability | Occlusion sensitivity implementation to replace unavailable Grad-CAM. | 5 |
| Evaluation | Labelled corpus assembly, label permutation study, accuracy measurement. | 9 |
| Verification | Numerical parity testing, smoke tests, defect resolution. | 7 |
| Deployment | Static build, hosting configuration, release pipeline, runtime vendoring. | 6 |
| Documentation | Cost statement, funder deck, hosting assessment, technical notes. | 6 |
| Governance and commercial | Funder strategy, governance requirements, IP and entity framing. | 5 |
| **Total** | | **66** |

*The workstream split above describes the scope as delivered across a full engagement, and totals 66 engineer days on that basis. Section 2 records the effort actually charged in this statement, which differs between scenarios because the rate basis differs. The scope delivered is the same in both.*

# 3. Infrastructure, data and API consumption

*MODELLED. Replace quantities with provider billing data before external issue.*

| Item | Quantity | Rate | Amount |
|---|---|---|---|
| Agentic development and code generation API consumption | 8.0 M tokens | £2.10 / M | £17 |
| Model training and evaluation compute | 10 GPU hours | £2.40 / hr | £24 |
| Dataset acquisition and transfer | 1 lot | £60 | £60 |
| Object storage and backup for artefacts | 6 months | £8 / month | £48 |
| Application hosting and content delivery | 6 months | nil (free tier) | £0 |
| Development tooling licences | 1 lot | £95 | £95 |
| Domain, TLS and identity services | 1 lot | £45 | £45 |
| **Total infrastructure and consumption** | | | **£289** |

*Planned allocations not yet drawn, including any supercomputing allocation referred to in the funding strategy, are excluded and will be reported against the period in which they are consumed.*

# Invoice

INV-2026-0002 | Date: 15 October 2026

**From**

GRATICULE INSTRUMENTS LTD

**Reference**

Applied AI platform development: model engineering, evaluation, interface and deployment

| Description | Qty | Unit | Amount |
|---|---|---|---|
| Design, build, verification and evaluation of the applied AI platform, delivered as a working prototype with browser inference, evaluation harness and technical documentation (lot). | 1 | £8,620 | £8,620 |
| Infrastructure, dataset acquisition and third-party API consumption incurred in delivery (lot). | 1 | £289 | £289 |
| **Subtotal** | | | **£8,909** |
| VAT (not applicable, entity not yet VAT registered) | | | £0 |
| **Total due** | | | **£8,909** |

**Purpose of this statement**

This schedule records development expenditure incurred on the platform to date. It is prepared for internal capitalisation and funder disclosure, and is not a demand for payment.

*Prepared by the technical function of GRATICULE INSTRUMENTS LTD. Figures marked MEASURED are reproducible from the delivered repository. Figures marked MODELLED are a planning schedule and are subject to confirmation.*